



SOMALIA: ECONOMICS OF WATER SUMMARY REPORT

Digging for Data

**Towards Understanding Water as a Limiting or
Enabling Factor for Socioeconomic Growth in Somalia**

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Towards Understanding Water as a Limiting or Enabling Factor for Socioeconomic Growth in Somalia

A summary report, along with a suite of six reports, accompanying the 'Somalia: Economics of Water' Report

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Abbreviations and Acronyms

AF	Additional Financing (World Bank)	PPP	Public-Private Partnership
AfDB	African Development Bank	SWC	Soil and Water Conservation
BRA	Benadir Regional Administration	SRTM	Shuttle Radar Topography Mission
CAPEX	Capital Expenditure	SWALIM	Somalia Water and Land Information Management (FAO)
ENSO	El Niño Southern Oscillation	ToR	Terms of Reference
ET	Evapotranspiration	UNICEF	United Nations Children's Fund
FAO	Food and Agriculture Organization	WALP	World Bank-funded Water for Agro-Pastoralist Livelihood Pilot in Somalia
FAOSTAT	Food and Agriculture Organization Corporate Statistical Database	WASH	Water, Sanitation, and Hygiene
FCV	Fragility, Conflict, and Violence	WaPOR	Water Productivity Open-Access Portal (FAO)
FSNAU	Food Security and Nutrition Analysis Unit	WB	World Bank
FGS	Federal Government of Somalia	WBG	World Bank Group
FMS	Federal Member State	WDA	Somali Water Development Agency
GDP	Gross Domestic Product	WDS	Water Delivery System
ha	Hectare	WFP	World Food Programme
IDP	Internally Displaced Person	WHO	World Health Organization
IMF	International Monetary Fund	WSC	Water and Soil Conservation
INGO	International Nongovernmental Organization		
km	Kilometer		
km ²	Square Kilometer		
l/cap/day	Liter Per Capita Per Day		
l/s/km ²	Liter Per Second Per Square Kilometer		
lcd	Liter Per Capita Per Day		
m ³	Cubic Meter		
m ³ /ha/year	Cubic Meter Per Hectare Per Year		
MCM	Million Cubic Meter		
MCM/yr	Million Cubic Meter Per Year		
microS/cm	Micro Siemens Per Centimeter		
mm/d	Millimeter Per Day		
mm/yr	Millimeter Per Year		
MoEWR	Ministry of Energy and Water Resources		
MoWD	Ministry of Water Development		
NDP	National Development Plan		
NGO	Nongovernmental Organization		
NWRS	National Water Resource Development Strategy		
ODA	Official Development Assistance		
OPEX	Operational Expenditure		
PET	Potential Evapotranspiration		

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Executive Summary

Water insecurity in Somalia continues to impact the lives of millions, inhibiting economic and human development. This report ‘Digging for Data: Towards Understanding Water as a Limiting or Enabling Factor for Socio-economic Growth in Somalia’, proposes ways to avert future crises by introducing untapped opportunities within the country. Six key messages emerged during the study.

Data are scarce. The report’s title was chosen as ‘Digging for Data’ because finding data on water in Somalia is almost as hard as digging for water in arid lands. Data are extremely limited, and whilst this study evolved to producing and analyzing far more data than was ever imagined, it also highlighted the need for ongoing studies.

A suite of technical studies was produced: a full Synthesis Report, a Groundwater Assessment and a Groundwater Quality Note, a Surface Water and Riverine Assessment, an Institutional Analysis and a Water Sectoral Analysis. They offer comprehensive water data for Somalia and can be downloaded from: the World Bank’s Water Global Practice website (<https://www.worldbank.org/en/topic/water>); the Bank’s Somalia website (<https://www.worldbank.org/en/country/somalia>); and, in 2022, from the Ministry of Energy and Water Resources’ (MoEWR) website (<https://moewr.gov.so>), currently under construction.

Is Somalia a water-scarce country? Not exactly, and the situation is not as hopeless as one would imagine. Analysis of water availability and expected use in 2030 at a national level shows that, except for the Benadir region, long-term renewable groundwater and surface runoff resources will be sufficient to meet water demands (human, livestock, and agriculture) without significantly affecting environmental flows—provided water is allocated to the most productive needs and managed efficiently. This is particularly important to overcome localized acute water scarcity at the district level and chronic vulnerability to variable rainfall at the community and household level, negatively affecting people’s health, education, and long-term human development outcomes, particularly for women and vulnerable groups.

Water insecurity creates economic and human development challenges through complex, multiple pathways (see figure on page viii). Water insecurity in Somalia, including exposure to ever more frequent floods and droughts, is amplified by environmental degradation and climate change, both of which can make rainfall more erratic, while reducing the ability of watersheds to naturally store, treat, and slowly release water for productive use and human consumption. The challenges of achieving Water for All has been further exacerbated by the global economic fall-out of COVID-19, resulting in poor macro-

economic performance which has led to low domestic revenue collection, capital expenditure, and competing sectoral priorities in turn reducing water sector spending. Sector financing constraints, combined with rapidly growing needs fueled by forced displacement and urbanization, further strain the state's ability to maintain inclusive and sustainable water services, amplifying social and gender inequalities. Breaking this cycle means building resilience to economic and climatic shocks through a more diverse and dynamic urban economy and a more secure rural economy, both of which need to be underpinned by affordable, inclusive, and reliable water services, itself a key pillar for improving human capital development.

Investing in water generates high returns.

Based on an analysis of recent rural water sector projects in Somalia, this study highlights how US\$1 million invested in rural water infrastructure and livelihood development activities is estimated to generate a discounted return of US\$5.85 million under baseline assumptions.¹ This study sets out a broad suite of potential investments for the short and medium terms for rural, peri-urban, and urban areas which would strengthen resilience at local levels, transform livelihoods and the economy, and enhance the availability of data, information, and knowledge on water and its links to Somalia's economy and social development.

Reinvent the riverine breadbasket through farmer-led irrigation. Much recent discussion has revolved around the demise of the Jubba and Shabelle riverine areas' irrigation infrastructure which, in years gone by, provided the water for Somalia's agricultural hub. Multiple reports and strategies discuss the need to rehabilitate the aged and dilapidated infrastructure. This study suggests a different path and proposes that, whilst rehabilitation of existing infrastructure is viable in the long term, the quicker, and cheaper, method of farmer-led irrigation development

should be embraced in the short to medium term. Where water sources are available nearby, farmer-led irrigation costs one-fourth as much on average as traditional irrigation and it is far easier to implement.

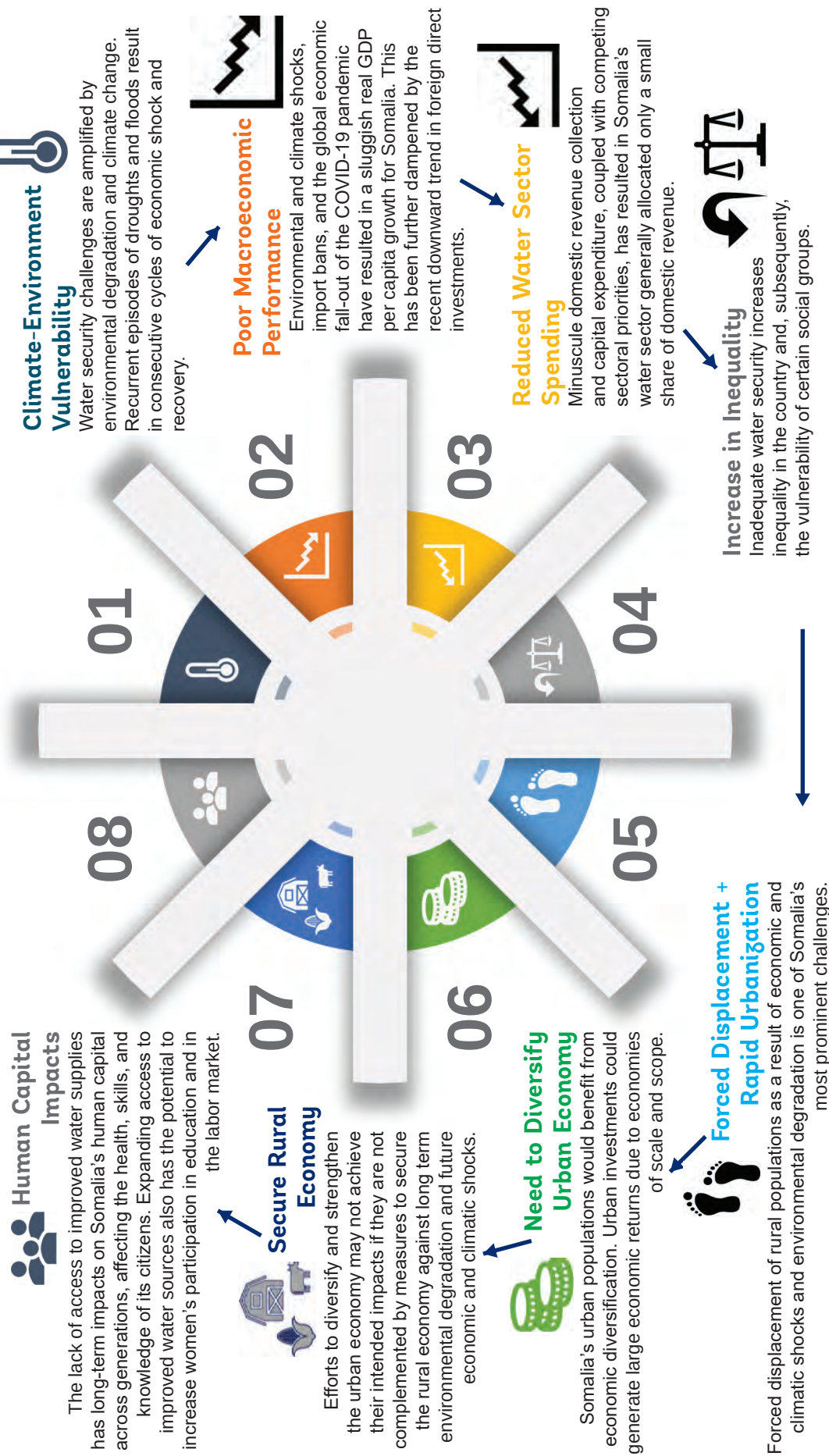
Understanding urban water supply and sanitation is an urgent analytical priority. The study found that the rural water sector is well understood and supported, and existing projects are scalable without much modification. The same cannot be said for urban water supply and sanitation, however. In the short term, analytical work should focus on understanding water and sanitation service delivery to households and firms in urban areas, unpacking the roles of emerging water utilities, and determining appropriate governance mechanisms. This will help provide services for growing urbanization and contribute to bridging gaps between rural and urban economies.

The final section of the study concludes with headline recommendations and detailed projects for consideration in the short and medium term. The study suggests that all water sector investments should be prioritized based on the twin objectives of (a) their potential to improve long-term water availability as a critical contribution to water security and climate change adaptation; and (b) the extent to which they contribute towards wider socioeconomic development, reducing inequalities, and creating economic and human development opportunities.

In July 2021, the Federal Government of Somalia published the National Water Resources Strategy (NWRS). A primary recommendation emerging from this study relates to the institutional arrangements in the water sector and it is an immediate foundational necessity for water governance to be underpinned by policy and legislative instruments. The NWRS includes a roadmap of flagship projects, and the concluding section and tables in this report align to those.

1. Somalia Water for Agro-Pastoral Productivity and Resilience Project Economic Analysis.

Headline Messages



Introduction

The report presents a high-level summary of research findings and policy recommendations from the ‘Somalia Economics of Water: Digging for Data’ study. The study’s objective was to analyze key pathways through which the availability, or lack, of water services can impact economic outcomes, resilience, and fragility in Somalia.

Three research questions were formulated:

- How much are water resources availability and quality of access to water services limiting factors for human and economic development in Somalia?
- What are the socioeconomic impacts of underinvestment as per consumptive water uses (with emphasis on water services for rural and urban areas and water for livestock and agriculture)?
- How could water be better harnessed in the future?

The ‘Somalia Economics of Water’ study was carried out under the Somalia Water Platform Program, which is managed by the World Bank Water Global Practice and financed by the Global Water and Sanitation Program. The Platform was launched in 2019 to provide a mechanism for dialogue and coordination on the water agenda with the World Bank Global Practices, thereby informing the World Bank on water-related engagements. Initial outputs of the Water Platform in Somalia included: (a) a sector note identifying challenges and opportunities for engagement in the sector; (b) service delivery model development as the water sector attempts to graduate from humani-

tarian responses to country-led development; and (c) a first analysis on the aspects of water economics in Somalia. The study produced a full Synthesis Report, ‘The Economics of Water: Digging for Data: Towards Understanding Water as a Limiting or Enabling Factor for Socioeconomic Growth in Somalia’, and a suite of complementing technical reports including a Groundwater Assessment and Groundwater Quality Note, a Surface Water and Riverine Assessment, an Institutional Analysis, and a full Water Sector Analysis. The full Synthesis Report and suite of technical studies are available on: the Ministry of Energy and Water Resources’ (MoEWR) website (<https://moewr.gov.so>); the World Bank’s Water Global Practice website (<https://www.worldbank.org/en/topic/water>); and the World Bank’s Somalia website (<https://www.worldbank.org/en/country/somalia>).

This summary report is organized as follows. **Section 1** describes Somalia’s water resources and how they are changing. **Section 2** describes water usage in Somalia—both currently and expected future demand. Somalia’s complex and unique institutional framework for managing water resources and providing water to users are discussed in **Section 3**. Based on the study’s analysis of available resources and future demand, **Section 4** lists a range of potential options for investing in water supply and management. The summary concludes with recommendations for investments, policy reforms, and additional research.

SECTION 1

What are Somalia's Water Resources? How are They Changing?

This section discusses Somalia's water resources and analyzes how they are changing with time. It further answers the question: Is Somalia a water scarce country or not? Groundwater is the primary source of water in Somalia and although sufficient water is available, on average, across the entire country, water quality is often low. The main groundwater quality issues are salinity, fluoride, and bacteriological contamination. Salinity affects both the use of water for drinking and for agriculture (Said 2021) while bacteriological contamination primarily affects the safe use of water for human consumption; even though there are not many systematic surveys, the studies that do exist report the problem of bacteriological contamination of dug wells, berkads, and hafirs (Berger 1985; EarthWater 1998; van Haren et al. 2017; WASH cluster Somalia 2017). Generally, there is significant spatial variation in terms of water resources, and one finds water scarcities when analyzing supply at the district and community level. There is large temporal variation due to seasonality, weather shocks, and long-term climate change.

1.1 Ground and Surface Water Sources

The total freshwater resources available in Somalia comprise the total volume of water that is available due to external flows, internal flows, runoff and groundwater renewal generated by precipitation

within the territory. Internally produced renewable water amounts to 7.8 cubic kilometers per year (km^3/yr) (FAO 2014), and Somalia also receives externally produced water from Ethiopia via transboundary aquifers and rivers and wadis.² These externally produced renewable water resources amount, in turn, to $8.2 \text{ km}^3/\text{yr}$. In other words, more than half of Somalia's water resources are produced from beyond its national borders (FAO 2014).

There are two perennial rivers in the country, the Shabelle and the Jubba, both of which enter the country from Ethiopia, where the bulk of the river catchments volumes are located. The Jubba flow is more than twice that of Shabelle. The Jubba river flow in Luuq, close to where it enters Somalia, is about $5.9 \text{ km}^3/\text{yr}$, and that volume is hardly reduced along the river flow towards the ocean at about $5 \text{ km}^3/\text{yr}$. The Jubba reduces by less than 20 percent from evapotranspiration, groundwater recharge, and utilization. In contrast, in Beledweyne district where the Shabelle river enters Somalia, the river flow is $2.4 \text{ km}^3/\text{yr}$, and about $1.5 \text{ km}^3/\text{yr}$ at the end point, an area of alluvial deposits, close to the Jubba river. In other words, the Shabelle flow almost halves before recharging groundwater within the basin. For most years, the Shabelle river is a closed basin, with all water consumed within the basin, except in occasional years

2. Wadis are valleys, ravines or canals that run mostly dry except during the rainy season.

when very wet conditions cause the river to overflow and discharge in the Jubba river and the ocean.

Groundwater is Somalia's largest natural water supply source and provides the main storage medium. Springs are also found in many places, but discharges are small and disappear downstream because of water withdrawals, infiltration, and evaporation. Large parts of Middle and Northern Somalia are underlain by low permeability sediments. However, deeper aquifers like the Auradu limestones in North Somalia or Quaternary sands are within drilling reach at a depth of between 100–400 m.

In addition, there are alluvial fills in basins such as the Nugaal valley, in the north of the Hargeisa depressions. There are also fan deposits along mountain ranges, alluvium along wadis and the Shabelle and Jubba rivers and coastal dunes. The alluvial fills, fan deposits, alluvium along the wadis, and two perennial rivers as well as coastal dunes all contain groundwater at shallow depths (<15 m). These are essentially perched aquifer systems overlying the deep regional aquifers. The combination of soft formation and shallow groundwater make them ideal for groundwater abstraction by means of shallow, manually-drilled boreholes, and low-cost water lifting devices.

The crystalline basement rocks in the Bay region in South Somalia and in the far north are not continuous aquifers; shallow, machine-drilled boreholes (30 to 60 m) drilled into the bedrock in these fracture zones may yield 0.25 to 1 l/s. The clay, siltstone, and gypsum deposits that cover large parts of Somalia such as the Sool, Sanaag, and Togdheer regions in the North where the Taleh Formation overlies the Auradu limestone are categorized here as nonproductive formations; however, they could yield water (often brackish exceeding the 3,500 microS/cm threshold).

Over the last century, deep motorized boreholes (50 to 400 m) have been drilled into limestone and sandstone aquifers and thick alluvial sand deposits. Generally, borehole depths increase from west to



Natural spring in Somaliland. (Photo courtesy: Stephen D'Alessandro)

east. Together with dug wells, they are the most frequent water points registered in Somalia. The average number of users is 385 people and 3,200 animals per borehole. Water from deep boreholes is more suitable for urban and industrial water supply, while cheaper systems that tap shallow aquifers, or water in sand dams, are preferred for rural and small-town water supply and livestock water. Also, water quality is why groundwater is preferred for human consumption over surface or polluted water in shallow dug wells.

Ground and surface water resources are, however, not distributed evenly across the country, due to regional differences in climate. In the southern part and the far northwest of the country, a hot semi-arid climate and a tropical savannah climate exist (Rubel and Kottek 2010). Rainfall varies with the region, ranging from 700 mm/yr in the south to less than 100 mm/yr in the northeast.



Borehole drilling in Somaliland. (Photo courtesy: Ahmed Mohamed Hassan)

1.2 Water Availability in the Face of Short-Term Weather Shocks and Long-Term Climate Change

The availability of water is not constant over time in Somalia. Seasonal rains and periodic droughts and flooding determine short-term water availability, especially in areas distant from the perennial rivers. Meanwhile, climate change is changing long-term trends in rainfall, temperature, and evaporation. These factors create the need to store water over time and provide a bridge during dry periods.

First the annual cycle of rainy and dry periods creates ephemeral rivers and seasonal water availability. Rains are mainly confined to the gu season (monsoon-like rains from March to June, often leading to flash floods) and the deyr season (starting in late September, ending in early December, and varying from place to place across the country). Runoff in wadis or toggas (seasonal rivers) persists for some hours to several days after rainfall. Overland flow arises when soils are waterlogged or water repellent or when rains are intense, notably if daily rainfall exceeds 20 mm/d (Basnyat 2007). This overland flow becomes concentrated and accumulated in low-lying areas and wadis.

In addition to annual rainfall seasonality, Somalia is subject to periodic droughts and years with excessive rainfall. In Somalia, mild droughts are expected to occur every three and two years for the gu and deyr seasons, respectively; moderate droughts are expected every 11 and 12 years; and severe droughts

every 28 years, for both seasons; extreme drought can be expected once every 37 and 110 years for the gu and deyr seasons, respectively (Shilenje and Ongoma 2014).

There is hardly any natural surface water storage in Somalia. Water can be diverted from temporary streams into masonry reservoirs, such as berkads (basin ponds) and hafirs or passively stored in wadi sediments behind sand dams and underground dams. Hafirs and berkads are typically fed by manageable runoff from upstream catchments of 2–3 km² (Basnyat 2007). During the dry season, reservoirs such as open water surfaces, but also hafirs and berkads may lose about 1 m of water from evaporation, which is a substantial part of their storage volume given that their depth is only a few meters.

The reliability of berkads and hafirs is not very high, as perceived by the respondents in the Somalia Water and Land Information Management (SWALIM) water points database. These systems fail to bridge the dry season during moderate droughts. Also, many shallow traditional dug wells, reaching only the upper shallow groundwater, will run dry during droughts. Boreholes that tap water which is well below the static groundwater, large sand dams, and underground dams are meant to be the best options to cope with the effects of moderate and severe drought. Sand dams and other managed aquifer recharge techniques can increase groundwater storage, which does not suffer from such high evaporative losses.

Climate change increases the need for improved storage solutions. While total rainfall volumes may remain the same, or even slightly increase (FAO and WB 2018), the variability is expected to increase. Rainfall is predicted to come down in heavier events, hence floods are likely to become more frequent in the Jubba and Shabelle basins (Petersen and Gadain 2012; Tierney et al. 2015). Droughts are also likely to become more frequent as an outcome of greater rainfall variability (FAO and WB 2018).



A sand dam in Somaliland. (Photo courtesy: Chantal Richey)

1.3 Social Consequences of Water Shocks

Lives and livelihoods are shaped by Somalia's uncertain water supplies. Water-related shocks in Somalia have contributed to poverty, displacement, biodiversity loss, and conflict. Extensive floods also result in significant crop losses and forced displacement and make roads inaccessible, which also affects the transport and marketing of agricultural outputs from unaffected areas; whereas droughts pose a risk to agriculture, floods can cripple crop and livestock production. Areas without alternative, decentralized sources of water are likelier to be classified as being food insecure when compared to those with ground-water resources.

Recurrent drought and flood events have had severe humanitarian consequences. In the early 1990s, 300,000 people died from hunger because of multi-annual droughts. Due to floods in 1997 and 1998, around 1,400 casualties were estimated and another 1 million were indirectly affected; more than 60,000 ha of crops and farmland were lost. It has been difficult to develop resilience to these frequent extremes.



Water scooping in Lower Shabelle. (Photo courtesy: Local media, Jowhar)

Variability and environmental shocks in Somalia manifest themselves through the movement, migration, and displacement of people, both as nomadic pastoralists and as internally displaced peoples (IDPs). Shocks to the rural economy push people into peri-urban pockets of poverty as well as IDP settlements. Health impacts for rural households are also significant due to the number of people with unimproved services, with an estimated 2 million people, or 22 percent, of the rural population currently relying on water from unimproved sources.



SECTION 2

How Water is Used in Somalia and How This is Changing

Section 2 describes who uses water in Somalia and what water is used for. Understanding of water contributions to the economy, from positions of sectoral outputs, international trade, and human capital is deepened, and the economic disruptions of the 2016–17 drought are presented. Scenarios for medium and long term water demand are touched upon, and the section concludes with benchmarking WASH in Somalia against other countries which contain arid and semi-arid zones, or which are located in the Horn of Africa.

2.1 Who Uses Water and for What is Water Used?

Inequality along income, gender, spatial, and other dimensions is critical to understanding water usage in Somalia. Inequalities in access to water supplies and sanitation between rural and urban populations, between IDP and non-IDP populations, and between men and women in Somalia are striking. It's noteworthy, for example, that Somalia is the only country in the world where there is a rural-urban gap of more than 50 percentage points in access to a basic water supplies (WHO and UNICEF 2019).

A survey in Somaliland found that approximately 60 percent of households relied on adult women to collect water (UNICEF 2011). Women who travel away from their residences to obtain water can risk their personal safety, with gender-based violence

high in rural and nomadic areas in the country (MoE-WR 2021).

Urban areas have grown over the last decades, increasing the demand for water by households and businesses. It is estimated that approximately 54 percent of Somalia's population now live in urban areas (World Bank 2021a). In cities, piped networks, point-source vendors at boreholes and dug wells, as well as truckers and animal and human-drawn carts deliver water to consumers. Large cities close to drought-prone areas have also received waves of IDPs that put pressure on scarce land and water resources. This is especially the case in regions with large IDP settlements, where inequalities in water access by IDP and non-IDP populations are prevalent.

Much-needed economic diversification in urban



Filling water in Saaxo Mudud, Central Somalia. (Photo courtesy: Ahmed Mohamed Hassan)

areas is limited by water availability. Urban investments could generate large economic returns due to economies of scale and scope. The Ninth National Development Plan (NDP9) stresses the importance of achieving economic growth through nontraditional sectors, including by adding value to the livestock and crop production sectors. Water availability in urban areas is essential for agro-processing and livestock value chain exploitation. In the case of cities that rely on surface water from rivers, this will require forward-looking investments in managing upstream and downstream demand, variabilities in flow and water storage, including large scale infrastructure. In the case of groundwater, some cities are already reaching the limits of groundwater availability, while others still have considerable potential.

In rural areas, water sources tend to be used for multiple purposes, including drinking, for domestic use, and to water livestock. Traditionally, pastoralists moved themselves and their livestock to access year-round pastures and water sources. While this remains the case for some, migration to IDP settlements has become a complementary coping strategy, as has family splitting. Movement of rural people to IDP settlements tends to be a response to flood and drought events but some IDPs remain in settlements.

2.2 Water's Contribution to the Economy

Water (including its absence) contributes to economic output, international trade, human capital formation, and poverty in Somalia. It is not possible to quantify these contributions—data on amounts of water consumed by different economic sectors do not exist, and even supply-side gross domestic product (GDP) statistics are not compiled at present, thus preventing one from estimating the contribution of water-intensive activities (for example, livestock production) to Somalia's GDP.³ Nevertheless, one can trace out the channels through which water flows in the Somali economy.



A water truck in Garowe. (Photo courtesy: Ahmed Mohamed Hassan)

2.2.1 Sectoral Output

The economic contribution of water is seen most clearly in livestock, agriculture, and forestry. Somalia encompasses large areas suitable for livestock grazing, browsing, and fodder production; fertile alluvial soils for staple cereals, oil seeds, legumes, and horticulture crops; forests that provide prized gums and resins, as well as charcoal for cooking. Over 1 million agro-pastoral and nomadic households in Somalia rely on farming and livestock for their livelihoods and are, thus, particularly vulnerable to the impacts of meteorological and hydrological droughts (FGS 2018).

The waters of Somalia's two perennial rivers have supported livestock rearing, which historically was the mainstay of the economy and the source of wealth for

3. For examples of water resource accounting where data do exist, see Glenn-Marie Lange and Rashid Hassan, *The Economics of Water Management in Southern Africa* (Edward Elgar Publishing, 2007); and Nicholas Kilimani, Jan van Heerden, and Heinrich Bohlmann, "Water resource accounting for Uganda: Use and policy relevancy," *Water Policy* 18 (February 2016).



Irrigation channel in Jowhar being dredged. (Photo courtesy: Ahmed Mohamed Hassan)

most households. The national goat and sheep herd remains below its pre-civil war peak (FAOSTAT data reported in FAO and WB 2018). Boosting livestock production depends fundamentally on ensuring year-round availability of livestock feed. Policies or investments into fodder production have potential to reap considerable rewards, although there is need to balance short terms gains with long term sustainability, and to consider the contentious issue of land enclosures vis-à-vis open grazing.

For many decades—or centuries—agriculture has been practiced along floodplains of the Jubba and Shabelle rivers, producing food for both local consumption and export. Cereal production peaked in 1989, however, and declined steadily during the 2010s, reflecting challenges of inadequate water and transport infrastructure, persistent insecurity, weak regulatory and enabling institutions, and severe environmental degradation of rangelands and forested areas (FAO and WB 2018). Nevertheless, there is considerable potential to rebuild both subsistence and commercial agricultural production, which would support integration of rural and urban economies and reduce food import dependency. This would require improving soil and water conservation techniques in areas of rainfed agriculture.⁴ Stone bunds, terraces, and half-moons reduce overland runoff and erosion, promote infiltration, and increase soil moisture content (Liniger and Critchley 2007). Irrigation can expand crop production in areas with adequate sur-

face water, as irrigation water demands of about 0.5 to 1 l/s/ha (50 to 100 l/s/km²) far exceed renewable groundwater resources (0.03 to 0.63 l/s/km²).

Water contributes to the economy through forestry, although one must acknowledge that deforestation in Somalia has been significant over the past 50 years, with flood plain forests essentially gone (one of several factors contributing to the disruptions caused by flooding). Given the ongoing unfortunate dependence on charcoal as a source of energy, and importance of forests in soil conservation, agroforestry is an important, and underexplored, subsector.

2.2.2 International Trade

Water is inextricably linked to the goods and services that Somalia trades with the rest of the world. Animal, agricultural, and forestry products—all of which rely heavily on Somalia's water resources—have dominated recorded goods exports since the country's independence.⁵ Bananas and other fruit and vegetables were the largest exports during the 1960s; sheep, goats, cattle, and camels became the dominant exports starting in the 1970s and, at different times, sesame and other oil seeds, meat, and forestry products (gums, resins, and charcoal) have averaged as much as 5–10 percent of recorded goods exports (World Bank Group 2021b). There is scope to

4. Less than 5 percent of land in Somalia is considered suitable for cultivation, of which over 75 percent is suitable for rainfed agriculture, and the remaining 25 percent has potential for irrigation (FAO and WB 2018).

5. Data reported by Somalia's trade partners to the United Nations COMTRADE database show that these products accounted for more than half—and usually more than 80 percent—of recorded goods exported from Somalia each year between 1962 and 2018. As services exports are currently of similar magnitude as goods exports, these products' share of Somalia's total exports is smaller—the latest official national accounts statistics show that livestock has accounted for 26 percent of total exports of goods and services between 2018 and 2020 (National Bureau of Statistics 2021). These statistics do not include Somalia's informal and other small-scale, cross-border exports to neighboring countries in the Horn of Africa, which may exceed total recorded flows and much of which is believed to be livestock (World Bank 2021b).

increase exports of these products and by developing agro-processing industries to increase the economic value from their production—and thereby to increase water’s contribution to the Somali economy (FAO and WB 2018).

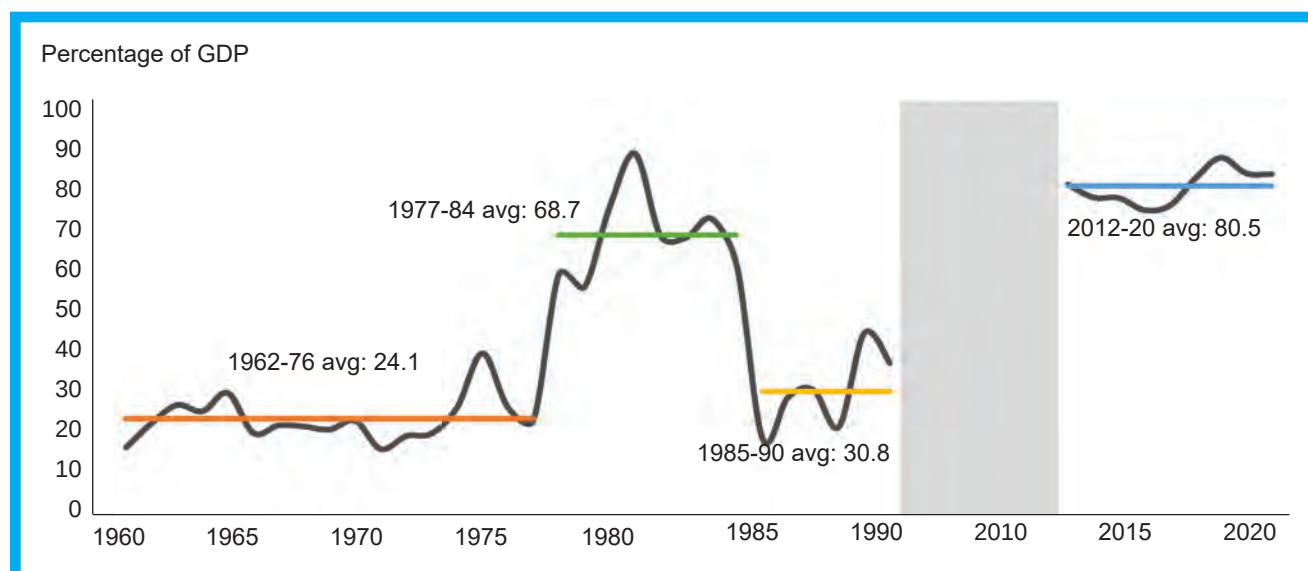
Water usage is also reflected in trends in the composition and level of Somalia’s imports. Displacement of the rural population to the cities and the destruction of irrigation systems caused by civil war, state fragmentation, and climatic shocks fundamentally transformed the commodity composition of imports. Since 1990, food, beverages, and tobacco have accounted for half of recorded goods imports, compared to 19 percent in the 1960s and 1970s, while the share represented by capital goods and transport equipment dropped to an average of 12 percent since 1990 from 40 percent before 1990 (World Bank 2021b). Total imports of goods and services now average around 80 percent of the GDP, as shown in Figure 1, compared to between 20 and 40 percent during most of the pre-1990 period. Somalia’s large trade deficit is financed largely by private remittances and official grants. As Somalia continues its trajectory out of fragility and rebuilds its economy, domestic food production should again become competitive with imports.

2.2.3 Human Capital

Water that is safe, reliable water and readily accessible—located in or close to the home—is essential for human health and, therefore, for labor productivity and human capital development. In both rural and urban areas of Somalia, diseases associated with poor water leads to reduced labor productivity, but the economic impacts are particularly acute in rural areas, where communities are constantly battling environmental and macroeconomic shocks instead of building more productive and resilient economies. Micronutrient deficiencies related to poor diets are widespread in Somalia and have far-reaching, long-term health consequences, including high rates of premature infant and maternal mortality and morbidity. Poor nutrition reduces labor productivity, with cascading impacts on income and poverty. It often leads to illness, a serious problem for poor households, because health costs can force them to tap savings, sell assets or go into debt, inevitably pushing them further into poverty.

Water also contributes to education. Improving access to education for girls, keeping them in school, and bringing them into secondary school education can enhance their opportunities to join the labor

Figure 1. Imports as a Share of GDP, 1962–2020



Sources: World Development Indicators (1960–1990) and National Bureau of Standards (2012–20). No data are available for 1991–2011.



Collecting water in Hudur Bakool. (Photo courtesy: Ahmed Mohamed Hassan)

market and contribute to economic development, and may also reduce Somali fertility rates. The water sector has an extremely important role to play to keep girls in school by ensuring that suitable WASH facilities are available, and that menstrual hygiene management is catered for. Given the paucity of information on education, including WASH in schools in Somalia, as reflected in international comparisons, a starting point is the collection of this data.

2.2.4 The 2016–17 Drought and Economic Disruptions

Water's contribution to Somalia's economy is perhaps most evident when water is absent, as during the drought of 2016–17. The Federal Government of Somalia's (FGS) post-disaster assessment (FGS 2018) estimated economic losses from the drought at US\$2.2 billion and damages at US\$1.0 billion. Losses and damages to irrigated and rainfed crops, livestock, and fisheries accounted for 59 percent of all the effects, while water and sanitation, environment, and transport make up 38 percent. An estimated 6.4 million of the total livestock population was lost, equiva-

lent to about US\$350 million. The drought displaced an additional 926,000 people to settlements in Mogadishu and other cities.

The frequency of drought (and more recently flood) episodes has repeatedly resulted in economic damages and losses, which tend to be met with a reactive disaster response. Managing drought and flood risks comprehensively will help develop resilience.

2.3 Scenarios for Medium- and Long-Term Water Demand

The 'Economics of Water' study developed a simple hydrological model to study the basins of Somalia's two perennial rivers. Comparing water demands or needs in 2030 for rural and urban population and livestock with the sustainably available water resources shows that water needs are below these critical values except for Benadir region around Mogadishu, where the high urban water demand in combination with a small and little water producing region makes the scarcity ratio jump above the absolute water scarcity ratio of 100 percent. The model also indicates that some locations in the Shabelle river basin would be excluded from growing cereals and perennials in 2050. Projected water demand numbers developed under the 'Somalia: Groundwater Assessment Technical Report' show that by 2030 water needs will range geographically from 5 m³/yr in Bari district to over 20 m³/yr in Gedo for livestock; urban water needs will range from approximately 8 m³/yr in Middle Shabelle to over 45 m³/yr in Benadir, and rural water needs from approximately 10 m³/yr in Middle Jubba to just under 30 m³/yr in Lower Shabelle.

Projections for 2035 show that long term average runoff would deliver enough water in Shabelle to sustain people and livestock, but water would be only available in the lower sections for a few months per year (that is, probably only during the wet season). Otherwise, the basin would be dry. By 2050, irrigated farming would be reduced, as compared to both 2020

and 2035, not because of land scarcity, but due to water scarcity. A reduction in inflows due to climate change, or with more water use upstream, is likely to render the situation more difficult to manage.

This analysis is at a regional level. If the modeling were carried out at district level, several districts would most likely exceed the scarcity limits, and if modeling were to consider the scale of settlements and their direct surroundings, water scarcity is expected in about 30 to 40 towns.

2.4 Benchmarking Comparisons of Water and Sanitation Data

Figures 2 and 3 (*overleaf*) benchmark Somalia against the 28 African countries which contain arid and semi-arid zones, or which are located in the Horn of Africa. The WASH statistics are drawn from the most recent Joint Monitoring Report (WHO and UNICEF 2021).

In terms of the main source of drinking water, 56 percent of Somalia's population use at least a basic service. This figure is within 6 percentage points of all other countries in the Horn of Africa apart from Djibouti (Figure 2a). The North African countries plus

Mauritania, Nigeria, Senegal, and Mali have considerably higher figures, as do South Africa, Namibia, and Botswana. Analysis of rural drinking water data presented in Figure 2b shows that the proportion of the population using limited source (requiring more than 30 minutes of collection time), is particularly high in Somalia at 37 percent, exceeded only by South Sudan (42 percent) and Burkina Faso (39 percent). This is illustrative of the rural-urban gap in water supplies.

At the national level, the use of basic sanitation in Somalia, by 39 percent of the population, is higher than most East and Southern Africa countries apart from Djibouti, South Africa, and Botswana (Figure 3a). However, there is considerable variation between countries overall. Comparing the use of basic sanitation between rural and urban areas, Figure 3b shows that there is a difference of 19 percentage points between rural and urban Somalis, with rural dwellers less likely to use at least basic sanitation. In three countries (Mali, Mauritania, and Sudan), use of at least basic sanitation is higher in rural than in urban areas. However, the rural gap with respect to sanitation is larger than Somalia's in most countries (apart from Zimbabwe, Kenya, and Uganda).



Staff of the Ministry of Environment, Agriculture, and Climate Change at a newly constructed water trough in Xamxama, Puntland. (Photo courtesy: Chantal Richey)

Figure 2. Comparison of Select Drinking Water Data between African Countries Containing Arid Regions and the Horn of Africa

(2a) Proportion of the total population using a main drinking water supply that is at least basic (%)



(2b) Proportion of the rural population using a limited source (takes more than 30 minutes) for their main drinking water supply (%)

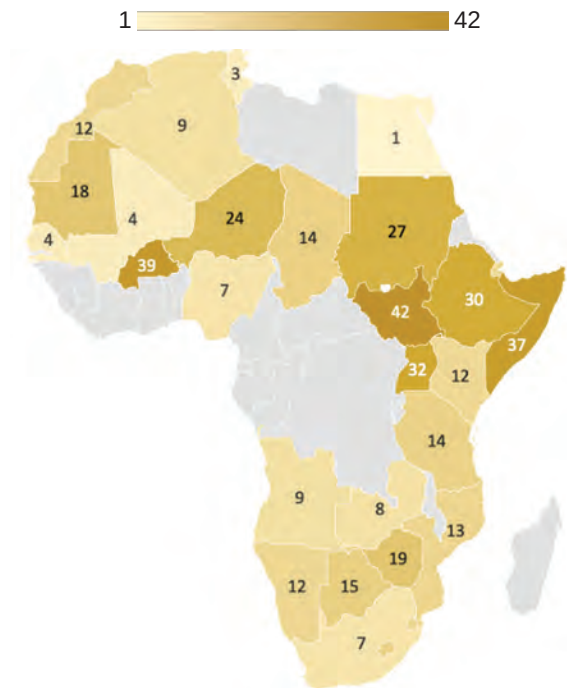
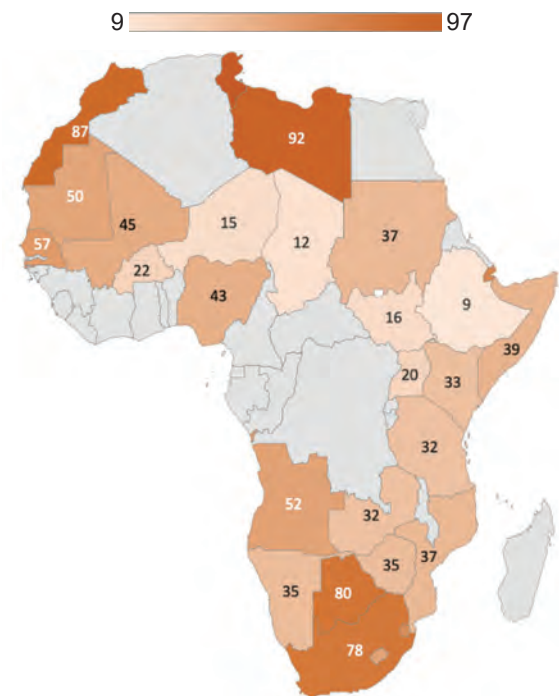
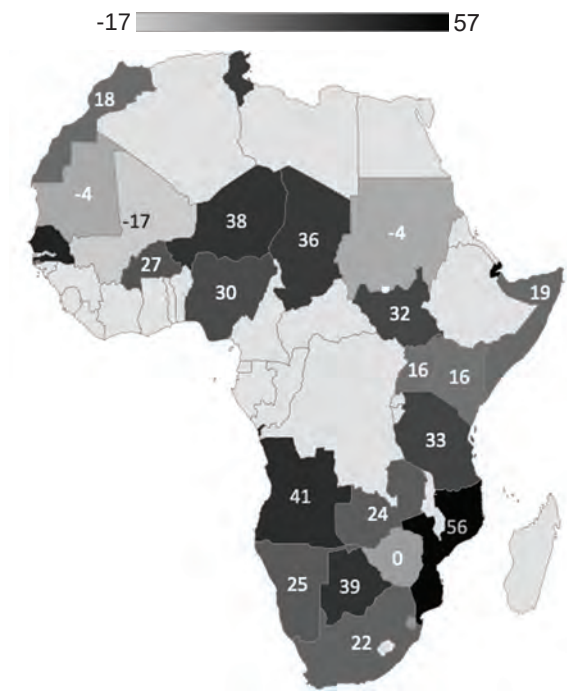


Figure 3. Comparison of Select Sanitation Data between African Countries Containing Arid Regions and the Horn of Africa

(3a) Proportion of the total population using at least basic sanitation (%)



(3b) Difference between rural and urban use of basic sanitation in percentage points (+ shows higher access in urban areas)



SECTION 3

How is Water Managed and Governed?

Although the Somali water sector has, in theory, many features of a comprehensive multigovernance framework, in practice, water service provision is minimally regulated and is currently dominated by international partners, development financing institutions, the private sector, and households themselves. This fragmented approach to service delivery comes along with challenges, particularly the coordination among the various actors, which is inadequate. The failure to translate policies (of which there are very few good quality ones) into action has led to the partial neglect of many intrinsic functions of a functional water governance framework—for instance, regulation of the quality of water supply services, setting optimal tariffs and collection of revenues, operation and maintenance of water services, and ensuring equity in service delivery.

3.1 Current Framework of Laws and Policies

The Xeer, Somalia's traditional law system, is an unwritten code of conduct that has governed its clan-based society over centuries. It includes provisions that deal with the management of water, pastoral land, forests, frankincense, and marine activity. The Xeer has specific rules for water sharing and queuing, especially during times of water scarcity, and settling water-sharing disputes between hostile clans.⁶

In the latter half of the 2010s, after a prolonged

period of state collapse, Somalia started developing the constitutional framework needed to anchor water sector institutions, thus establishing a rudimentary multilevel water governance framework. Key legislations are:

- The Water Act of 2011 is the core legal instrument of the Somali water sector. It made the Ministry of Water Development (MoWD) the sole custodian of the development and management of water resources. However, due to legislative issues surrounding the law's structure, as of early 2021, the provisions of the Water Act are yet to be implemented.
- Somalia's Provisional Constitution (2012) has specific articles related to the usage and management of water in Chapters 2 (Articles 25-1, 27-1), Chapter 3 (44, 45-1), and Chapter 5 (54(1a)).

In terms of policy landscape, the National Water Resource Development Strategy (NWRS) of 2021 contains detailed directives to operationalize the water sector's governance framework as laid out in the Water Act of 2011. It is the latest effort by the government to modernize institutions and restore services. The NWRS's provisions contribute to the strategic plan for activities required to address challenges in the water sector in the short- and long-term.

The Water, Sanitation and Hygiene (WASH) Sector

6. Somali Customary Law and Traditional Economy, Puntland Development Research Center, 2003.

Policy, 2019, encapsulates several policy measures related to water service delivery—urban and rural water supply, community-level water resource management, and water quality. Somaliland, Puntland, and Jubbland have developed state-level water policies and have made some progress towards their implementation. The Livestock Sector Development Strategy, the National Climate Change Policy, and the National Adaptation Programme of Action directly or indirectly have a bearing on the water sector.

3.2 Public Sector Institutions and Capacity

The public sector consists of the following ministries at the Federal Government of Somalia (FGS) and Federal Member State (FMS) levels:

- The Ministry of Energy and Water Resources (MoEWR) is the federal ministry responsible for the management of energy and water resources at the federal level. Its mandate includes the formulation of policies and standard operating procedures, monitoring and technical support of the FMS concerning the water and energy sectors; and the delivery of sanitation services.⁷ Sanitation services are not currently coordinated by the MoEWR due to capacity constraints.
- The Department of Water, one of the many functional departments of the MoEWR, is responsible for the planning, management, and regulation of the water sector. It consists of four divisions:

(a) Water for Production; (b) Technical; (c) Water Development; and (d) Water Supply.

- Federal Member States: Each FMS—Puntland, Galmudug, Hirshabelle, South West State, and Jubbland—has a Ministry of Energy and Water Resources line ministry that implements policy and performs a range of de facto functions related to the implementation of water infrastructure development and rehabilitation projects.
- In Somaliland, the Ministry of Water Resources discharges functions similar to the MoEWR.

Public spending on new water infrastructure and responses to water-related climate shocks such as drought and floods is limited. Insufficient revenue mobilization is one fundamental constraint. Total domestic revenue collected by the FGS, Somaliland, and the five FMS was estimated at 10 percent of GDP in 2018—considerably lower than revenue mobilization in neighboring countries or in other fragile and conflict-affected states (World Bank Group 2020). Budget allocations to the water sector are minimal. The federal MoEWR has consistently received less than 1 percent of the FGS budget since 2014 (Table 1). Virtually all its spending goes to salaries and other employment costs. Given the severe budget constraints as well as competing priorities of security and state-building, allocation of the FGS budget to all capital investments is small—averaging only 3.5 percent of

7. MoEWR website. Retrieved on February 22, 2021: <https://moewr.gov.so/about-us/mandate/>.

Table 1. Key FGS Fiscal Indicators, 2014–20

	2014	2015	2016	2017	2018	2019	2020
FGS capital spending (US\$ million)	\$0.16	\$2.29	\$5.94	\$6.35	\$9.64	\$13.93	\$18.55
Share of total FGS spending	0.11%	1.7%	3.5%	2.6%	3.4%	4.4%	3.7%
Total MoEWR spending (US\$ million)	\$0.62	\$0.45	\$0.49	\$0.88	\$1.02	\$1.11	\$1.77
o/w personnel costs (US\$ million)	\$0.51	\$0.38	\$0.45	\$0.82	\$0.98	\$1.07	\$1.09
MoEWR's share of FGS spending	0.41%	0.33%	0.29%	0.36%	0.35%	0.35%	0.36%

Source: World Bank staff calculations using Ministry of Finance outturn data (www.mof.gov/fiscal).

total FGS spending between 2014 and 2020—and mostly financed by donors.

Financial constraints also limit public sector staffing. The shortfall of adequately trained personnel across all functional domains—technical, administration, information technology and finance—across the FMS water ministries was cited consistently by key informants for this study. Limited financial resources are a binding constraint to the recruitment and retention of qualified personnel, further exacerbating the public sector skill gap in the water governance framework. Key informants expressed their belief that the skills gap in the public water sector is driven not by the scarcity of qualified personnel in the country but by the significant differences in salaries offered by the public sector and private companies, nongovernmental organizations (NGOs), and international organizations. Competing for capable and experienced personnel is a critical challenge for the Somali public sector.

The infrastructure and equipment installed during the eras before the Somali Civil War are dilapidated and/or outdated because of neglect during the civil war years. As such, present FMS water ministries have limited infrastructure and equipment to carry out their functions effectively. This extends to ministry buildings, information technology infrastructure, logistics infrastructure, and technical equipment.

The fragmented nature of the water sector, the amorphous distinction of roles and responsibilities concerning policies, and human resource capacity constraints diminish the regulatory and coordinative capacity of federal and state ministries. Thus, the actions of various players in the Somalia water sector are poorly regulated. The lack of coordination is a barrier to the efficient use of available resources and financing. The dilution of the role of state-level water ministries is manifested in the absence of standardization in the process of securing approvals for the construction and operation of water resources



The dilapidated state of irrigation infrastructure in the riverine areas of Southern Somalia. (Photo courtesy: Ahmed Mohamed Hassan)

across the FMS. High monetary overheads associated with monitoring and evaluation functions of FMS water ministries are another obstacle to the efficient discharge of its mandated roles and responsibilities.

Though constitutional provisions mandate regular meetings between federal and state-level ministries to coordinate water management issues, most interactions are conducted on an ad hoc basis. The irregularity of such multilevel meetings results in the misalignment of priorities and needs assessments between the two levels of government.

The ‘Somalia Urbanization Review’ (World Bank Group 2020) cites urban centers as Somalia’s most powerful and promising anchors of future development. However, contested, fragmented, and ineffective municipal governance is a key issue and the review recommends that immediate steps are taken to clearly demarcate responsibilities between national, state, and district or municipal authorities. This scenario plays out in the water sector, where service delivery remains largely unmanaged by municipalities. Clan alliances also exist within cities, which can weaken social cohesion and service delivery. Alas, demographic data on Somali cities remain inadequate and not all city boundaries are clear, presenting challenges for infrastructure development planning.

3.3 Water Services

These multiple levels of public sector institutions are complemented by a vibrant private sector.⁸ Water services across the country are delivered by public, private, and international institutions, as well as by households themselves, and are largely unregulated.

Private operators play a critical role in service delivery, filling gaps in the public provision of water services. Private water providers fall into three broad categories:

- (a) Aspiring water utilities, which are relatively sophisticated private businesses that operate boreholes and supply piped water connections to large parts of many primary/secondary cities.⁹
- (b) Small-scale providers, a fragmented network, ranging from small businesses to individual entrepreneurs, which serve smaller sections of urban and peri-urban areas.
- (c) Private entities that provide water services in territories under the control of Al-Shabaab. There is very limited information about the operations of these businesses.

Though many public-private partnerships (PPPs), loosely defined, are in place between FMS ministries and private providers, the limited oversight and monitoring capacity of FMS' MoEWR often precludes the public dimension in such partnerships. Even though

the policy framework vests regulatory authority (concerning tariffs and operational standards) with the FMS MoEWRs, their inchoate nature and limited capacities prevent them from comprehensively exercising their regulatory functions. The limited regulation of the private water sector is frequently cited as the main reason for high water prices across Somalia (discussed in more detail below). While FMS MoEWRs can keep track of relatively large quasi water utilities and monitor the entry of new entrants in this segment, the multitude of individual water entrepreneurs in every FMS precludes their efficient tracking and regulation.

Village- and community-level water committees play a central role in water resource management. Often, they require extensive capacity-building before they can independently manage the operations of their respective water points. Key informant interviews touched upon plug-and-play type water resource development projects, which often fail due to the inability of the hastily trained water committees to sustainably manage the operations of the respective water point. Rent-seeking behavior on the part of powerful community members in the water committees, limited enforcement of the roles and responsibilities post-hand-over, and a lack of ongoing external support for water committees leads to irregularities in the revenue collection process.

In addition, United Nations (UN) agencies, donors, international agencies and NGOs also play a significant role. The international development community has for many years provided Somalia with vital humanitarian emergency support by establishing

8. This assessment is based on the principles of good water governance defined in the AfDB's Water Sector Governance framework. These principles include equity, efficiency, participation, decentralization, integration, transparency, and accountability.

9. Number of large operators: nine in Hirshabelle, three in Jubbaland (excluding an undefined number in Lower Jubba and Gedo regions), and three in Puntland. Although Galmudug records having 31 operators, it is unlikely that all of these can be classified as large operators.



A shallow well in Hudur Bakool. (Photo courtesy: Ahmed Mohamed Hassan)

systems to fund, plan, and deliver assistance across the country, often in parallel with the government's structures and efforts. The Somali WASH cluster is a collaboration of local and international NGOs, government departments, and UN agencies involved in the WASH sector, and is led by the United Nations Children's Fund (UNICEF). The main objective of the cluster is to strengthen the capacity for emergency preparedness and disaster risk reduction through efficient and transparent dissemination of data and information. The cluster mechanism is appropriate for humanitarian response coordination during and post crises. With stability, however, sector oversight is the remit of the MoEWR and the cluster system functions should be taken over by government.

With growing political confidence, security, and stability in Somalia, there is a need for the support that is provided by the UN and other international organizations and NGOs, to transition from emergency humanitarian support towards long-term programmatic support. The institutional arrangements that the federal and state governments provide should be utilized during programmatic implementation to provide oversight and coordination, and to promote national ownership and long-lasting results within the water and development sector.

3.4 Gender Parity and Social Inclusion

Gender issues in Somalia are multilayered and complex. They are particularly relevant for the rural water sector where at the community level, while one or two positions are reserved for women in village water committees, customary law and cultural norms undermine women's input into management and decision making.

As seen in other countries, when women are not included in the planning process, their concerns—such as where a water source should be constructed—are not usually considered (Khandker et al. 2020). Research in other countries also shows that the active



Rainwater harvesting in Puntland. (Photo courtesy: Ahmed Mohamed Hassan)

participation of women in local water supply management can support mobilizing finance and fee collection and enhance sustainability (Lundqvist 1999; van Wijk-Sijbesma 1998), while if they participate in water user associations it can increase their social standing (Khandker et al. 2020).

Women have important roles to play in water management and need to be very well supported. Though women in Somalia are often consulted about the community management of water resources, they have been traditionally excluded from the final decision-making process. While water committees may reserve one to two perfunctory spots for women, in the Xeer system, women are excluded from being part of the committee of elders and are not allowed to act as advocates for others.

These gender disparities weaken effective water governance in Somalia, and it is essential that reforms to water governance and investments include women. At the national level, gender inequality is also evident in women's low participation in leadership and decision-making roles. Although data on women's participation in Somalia water sector institutions are not available, anecdotal evidence indicates that most decision makers are men, and that women rarely play a central role in defining water policies, programs, and laws. Women's low participation in de-

cision making is influenced by religious and cultural factors, and also by a lack of training and education opportunities in the water sector. An overwhelming majority of female respondents who participated in a 2020 survey indicated that they want more opportunities to participate in decision making (92 percent) and indicated that they would have more access to a decision maker if that person was a woman (86.6 percent) (World Bank 2021c).

3.5 Water Prices and Market Regulation

An analysis of water prices collected by the Food and Agriculture Organization (FAO) at weekly intervals over five years between 2015 and 2019 reveals two key issues. First, even though water prices across Somalia are significantly higher than in surrounding countries, there is substantial variation across regions within the country (Figure 4). While a spike in prices in 2017 corresponds with the onset of drought

conditions, some regions, including Sool, witnessed more drastic increases in prices. Second, while the divergence in water prices across regions is also evident in the plot of the weekly price of water (Figure 5), some regions, including Awdal, Waqooyi Galbeed, and Togdheer seem to experience far lesser variation in prices over time.

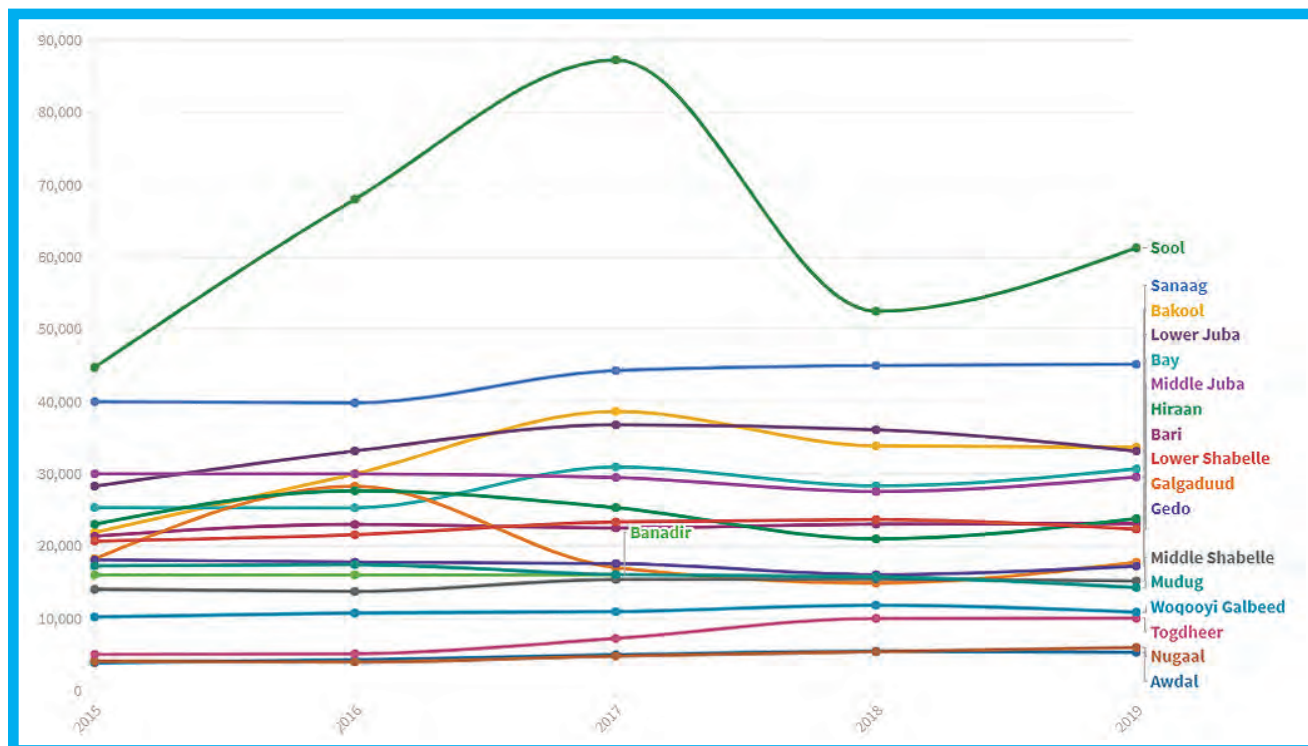
The ‘Somalia Economics of Water’ study’s analysis of price trends may be suggestive but is hardly conclusive. Deeper analysis is needed to inform policy development and investment choices. This analysis might look at the interplay between market structure (number of suppliers, barriers to entry, and so on), enterprise characteristics (ownership, size, cost structure, capital stock), locational factors (proximity to water resources, customers, conflict, and so on), the regulatory environment, and seasonality.

The National Water Resources Strategy 2021–2025 (NWRS) proposes regulations of water pricing and rules on competition to overcome economic inefficiencies that prevail in Somalia’s context of limited water resources. Insecurity and poorly specified property rights discourage investment, keeping scale low and unit prices high. The ‘aspiring utilities’ have market power and are able to charge prices well above costs in the absence of any effective government regulation. Since large-scale storage is largely nonexistent, weather shocks quickly become price shocks.



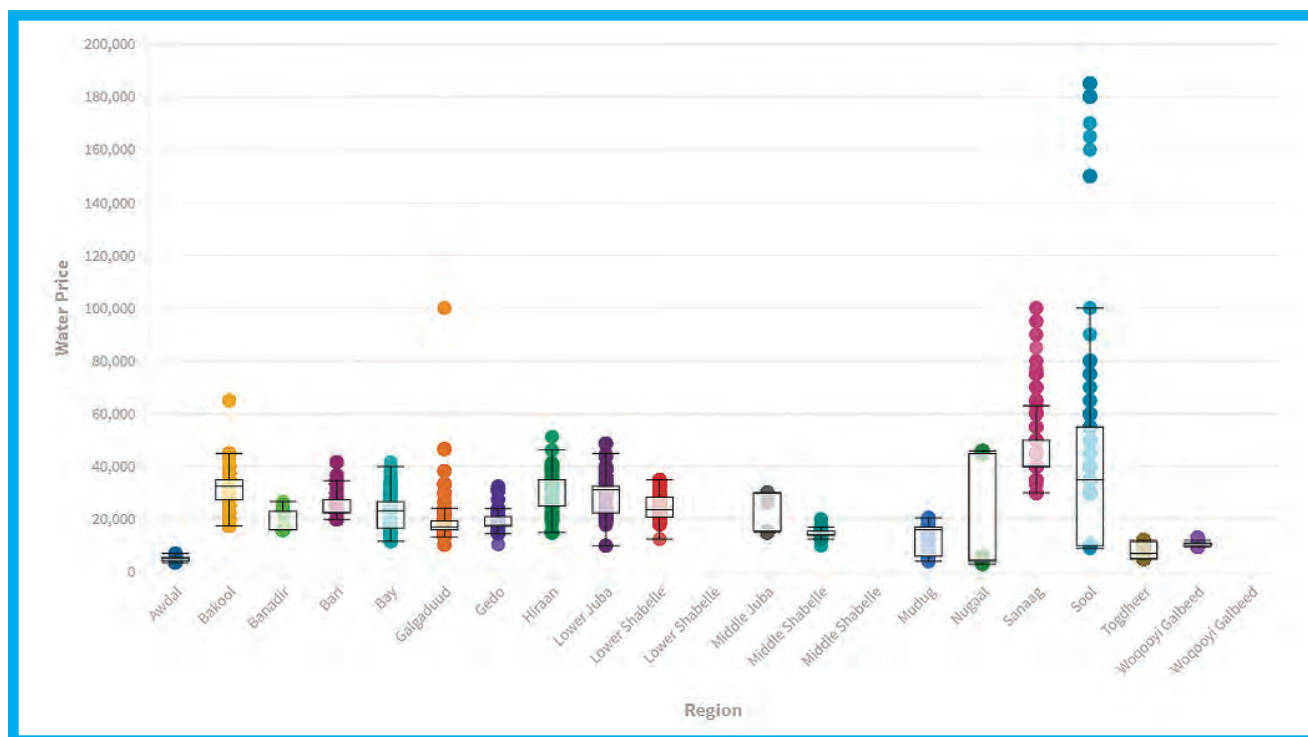
A wide variety of users gather at water points in Hudur Bakool. (Photos courtesy: Ahmed Mohamed Hassan)

**Figure 4. Water Price by Region—
Water Drum (200L): Average Annual Price (2015–2019)**



Source: Staff calculations using FAO data.

**Figure 5. Water Prices by Region—
Water Drum (200L): Weekly Prices (2015–2019)**



Source: FAO.

SECTION 4

What are the Investment Needs and Options?

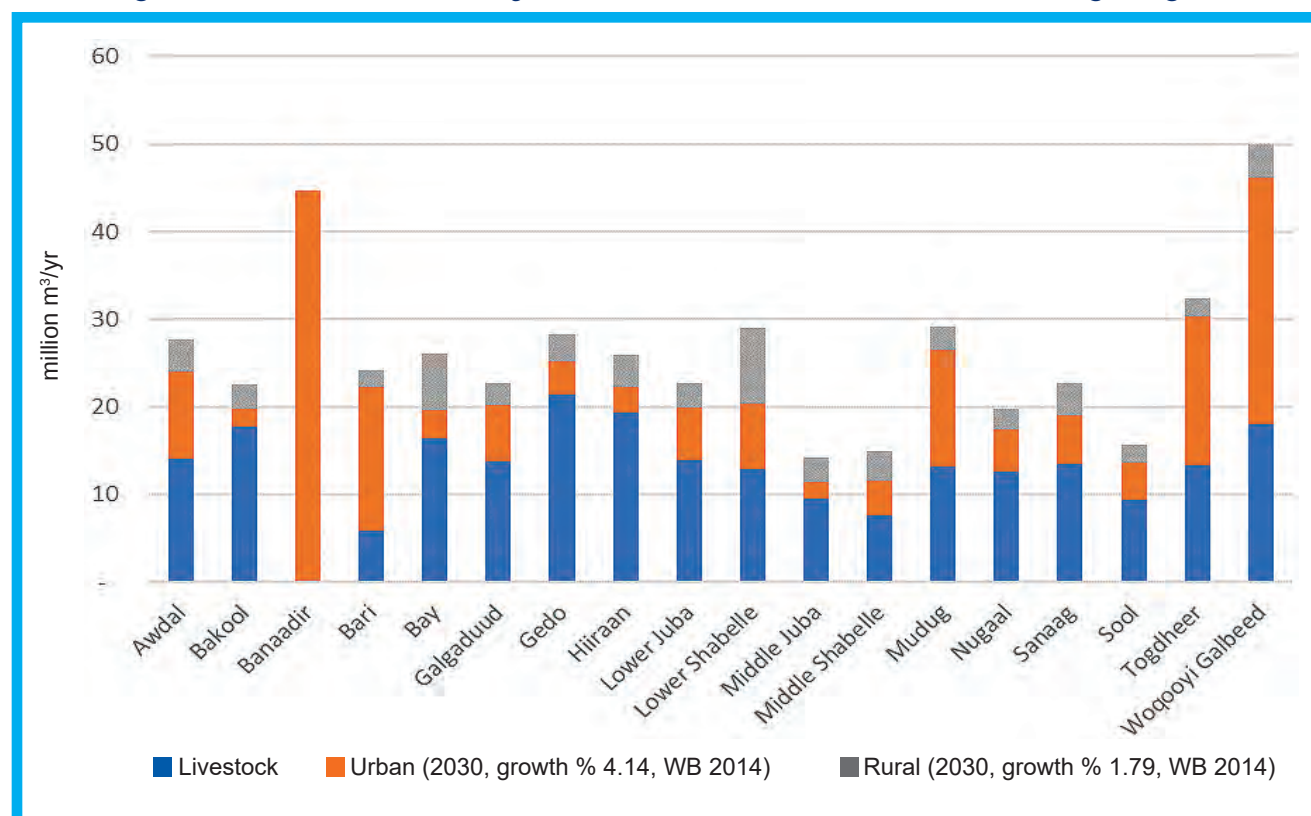
4.1 Gaps between Demand and Supply

The study estimates regional water demands in 2030 (Figure 6) and finds that, except for the Benadir region, water needs are below sustainably available water resources. However, water shortages still occur in certain districts, or at places with highly concentrated demand like towns or during drought.

In the Benadir region around Mogadishu, the high urban water demand in combination with a low pro-

duction means that water scarcity is very high. As noted in Section 1, at the scale of settlements and their direct surroundings, this study foresees water scarcity for about 30 to 40 towns. While boreholes are important for Somalia's drinking water supplies, high yielding boreholes are not abundant. This needs to be considered in relation to urban development and calls for diversification of water supply sources in some localities.

Figure 6. Sources of Projected Water Demand in 2030, by Region



Source: World Bank, Somalia: Groundwater Assessment Technical Report (2020).



A berkad in Somaliland. (Photo courtesy: Chantal Richey)

The amount of water assumed as being used today for household purposes in Somalia is estimated around 20 l/cap/day in rural areas and 40 in urban areas. One should consider that these usage levels reflect high poverty levels. Note further that water needs are based on annual figures. For specific dry season months or dry years, there will not necessarily be enough water available to meet needs. In such cases, improving water storage or access to stored water is essential. Dedicated water storage in shallow sand aquifers can increase well capacities and enhance climate resilience.

The simple model developed by the study indicates that, in 2050, there will not be sufficient water available to irrigate land that is potentially available in Shabelle basin for agriculture. There is a need to consider water allocation, appropriate river infrastructure, and improving water use efficiencies. There is also a need to promote regional development and win-win opportunities within the Horn of Africa.

4.2 Available Options

As described in earlier sections, groundwater is the principal water resource available in Somalia, and this section summarizes the pros and cons of the main technologies used for water supply. A cost-effectiveness analysis, described in detail in the full Synthesis Report, has been undertaken to show the relative contribution of each one of these water supply options to bridge the water gap, thus suggesting some criteria for the prioritization of investments.

Berkads are open-plastered cisterns dug in unconsolidated soil or hewn in bedrock. Storage capacities generally range from 150 to 500 m³. Hafirs, in turn, are created behind dams on gently sloping terrain, preferably on clayey soils to reduce infiltration losses. Storage capacities of these water harvesting systems range from 500 to 10,000 m³. Depending on rainfall, berkads and hafirs store water for approximately



An earth dam in Puntland. (Photo courtesy: Ahmed Mohamed Hasan)

two to five months after the end of the rainy season. Evaporation is high in these reservoirs (potentially 1,500 to 2,500 mm/yr) and leads to large losses. Just like the hand-dug wells, the berkads and hafirs are bacteriologically polluted and not safe as sources for drinking water supply. The estimated annual total cost per water produced is US\$1.51/m³.¹⁰

Another more recent intervention are sand dams and underground dams (subsurface dams) in combination with collector drains and wells. Sand dams are constructed in the alluvial beds of wadis (toggas/seasonal rivers). The storage capacities of the sand bodies behind these dams vary from 5,000 to 40,000 m³. This volume is available for use during the dry season (180 days) but is often augmented by recharge through groundwater inflow from the surrounding soils. The estimated annual total cost per water produced is US\$0.25/m³ with a handpump and US\$0.52 with a motor pump.

By its side, production capacities of hand-dug wells, in alluvial unconsolidated sands in tectonic depressions or along the wadis, is about 1 m³/hr or 5 m³/d depending on the type of pump. As the wells depend on shallow groundwater the water levels fluctuate strongly both seasonally and interannually. At the end of the dry season dug wells often run dry.

10. Statistics on water production costs presented in this subsection can be found in Table 8 of the full Synthesis Report.

However, water in most open traditional dug wells is compromised by the presence of pathogen microorganisms. Improved (lined and closed) dug wells with handpumps provide better quality water. The estimated annual total cost per water produced is US\$0.72/m³ including a handpump and US\$0.53 if a motor pump is used.

Subsurface dams and wadis provide important shallow groundwater supplies and may have further untapped potential. Better use of Somalia's groundwater resources presents an opportunity to improve resilience by relying on the buffering capacity offered by considerable storage reserve volumes. However, water quality does need to be considered, particularly salinity, fluoride, and microbiological contamination.

Groundwater use in Somalia could be further enhanced in certain areas with the introduction of relatively cheap manual drilling. Manually-drilled wells are a practical alternative for boreholes in alluvial soils and soft rock formation with a depth up to 40 meters, are lower cost than machine-drilled wells, and are also a good alternative for hand dug wells. Furthermore, manual drilled boreholes can penetrate deeper than excavated wells, and if properly sealed, they provide better protection from surface contamination. The estimated annual total cost per water produced is US\$0.58 m³ with a handpump and US\$0.46 with a motor pump.

Constructions such as sand dams and underground dams, already successfully applied under the Biyoole project, should be continued. Recharge ponds surrounded by shallow wells are also options for further investigations. Apparently, they already exist as van Haren et al. (2017) mention the presence of hafirs on permeable subsurfaces in Somaliland, where infiltrated water is captured by shallow wells.

The technology options presented above relate predominantly to rural and peri-urban water supply. Future analysis is required to build on options for urban water supply and sanitation.

SECTION 5

Recommendations Summary

Water insecurity negatively impacts human capital through multiple pathways (Figure 7) and is a deterrent to economic growth and social development. Conservation of water resources and aquatic ecosystems alongside environmental regeneration is a prerequisite to make the gains of social and economic progress sustainable. Investing in increasing improved access to water and sanitation is an essential driver of economic growth and human development.

The ‘Somalia Economics of Water’ study concluded that all water sector investments should be prioritized based on the twin objectives of (a) their potential to improve long-term water availability as a critical contribution to water security and climate change adaptation; and (b) the extent to which they contribute towards wider socioeconomic development, reduce inequalities, and create economic and human development opportunities. Somalia can use water access to improve coping measures and build resilience for rural communities through pathways which improve subsistence agriculture, create more jobs, and support urbanization.

5.1 Priorities for Policy Reforms and Institutional Strengthening

In July 2021, the Federal Government of Somalia (FGS) published the National Water Resources Strategy (NWRS). A first for Somalia, the NWRS describes

the critical economic and social importance of water in Somali society. It identifies opportunities to use access to water as a key stepping stone for socioeconomic development as well as an entry point to peace building. Managing Somalia’s water resources requires a strategic national approach that involves the engagement of key government actors, the private sector, civil society, as well as the support of a range of international cooperating partners and development financing institutions. The NWRS provides a suite of strategies, objectives, and actions for the water sector for 2021–2025, although some of the expected outcomes will only be realized in the longer term. The NWRS targets to unlock key actions and align with the Provisional Constitution (2012), Sustainable Development Goals, and sectoral policies and laws. The NDP9 will be the key driver for the NWRS over the next five to 10 years. The vision of the NWRS is: “Sustainable, equitable and secure water for national unity, growth and well-being, for all and in harmony with nature”. The NWRS includes a roadmap that sets out flagship projects numerically and the concluding tables in this report align to this NWRS roadmap.

With the finalization of the NWRS 2021, proposing and assignment of public sector responsibilities for federal, state, and local governments as set out in the governance frameworks should be expedited. Institutional arrangements are pivotal to the success

Recommendations

of any sectorwide reforms as they are the vehicles for implementation of policies and strategies. Providing clarity regarding mandates, roles, and responsibilities is important to ensure effectiveness and efficiency. This is particularly so when capacity and resources are constrained. The Somali institutional framework for water governance is evolving and, through the development of the NWRS strategy, progress has been made in gaining improved alignment between the FGS and FMS ministries. Building on this will be a process to formalize these arrangements, through policy and legislative instruments.

The reality of the need for continued external funding of the water sector should not be a barrier to strengthening long-term institutional and technical capacity, and development programs should be aligned with the NWRS and contribute to country system strengthening. Flood and drought risk management measures, both ex-post response and ex-

ante flood prevention and mitigation measures, need strengthening as well as coordination between the MoEWR and the National Disaster Risk Management actors. There is a need to manage drought and flood risks comprehensively in order to help develop resilience. Considering floods as a resource and opportunity should be encouraged, and Somalia should continue to invest in pathways to harness floods such as having sand and subsurface dams.

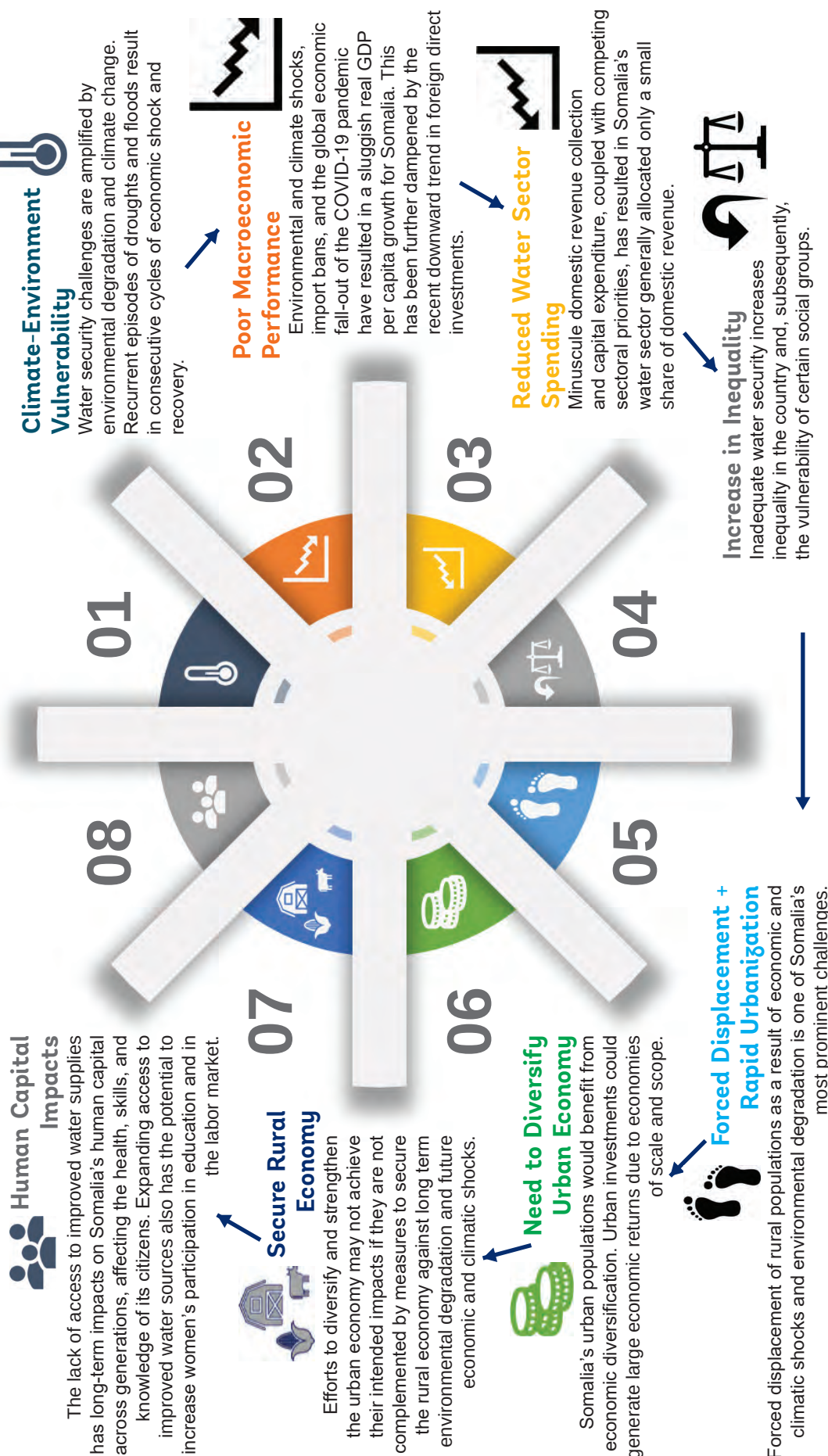
Before any sustainable large investment infrastructure is done, a policy intervention is needed relating to the defining of a clear urban service delivery model by the government, wherein the financing of large-scale urban water supply and sanitation infrastructure is contingent on the development of a service delivery model for urban water and sanitation supply. The simultaneous development of an Urban Water Sector Investment Plan is very important. This can comprise a consolidation of a series of City Summary Invest-



A small-scale water vendor in Hargeisa, Somaliland. (Photo courtesy: Chantal Richey)

Figure 7. Water Insecurity Negatively Impacts Human Capital through Multiple Pathways

Headline Messages



ment Plans. While there is an urgent need to ease the burden on existing water delivery systems in urban areas, efforts need to be mindful of the considerable commercial interests by those currently filling the gap in service provision.

5.2 Priorities for Physical Investments and Capacity Strengthening

Investing in water generates high returns and this study sets out a broad suite of potential investments for the short and medium terms for rural, peri-urban, and urban areas. Whilst domestic revenue mobilization and poor macroeconomic challenges persist, the reduced national water sector spending will continue to be a barrier for service delivery and the need for external funding remains essential. There is a need to channel external funding for service delivery in the sector, including investing in water infrastructure projects that can yield short-term returns. Such projects must support capacity strengthening, sustainability, and self-sufficiency. It is recommended that all investments into infrastructure fully integrate components related to building institutional and technical capacity in the water sector and insofar as is possible, particularly in rural Somalia, projects should be framed around community-driven development principles and sustainable land and water management.

Investments in water infrastructure need to reduce

inequalities, for example, between different groups such as IDP/non-IDP and different cities, and between urban and rural. Where feasible, investments should have the potential to expand access to water for income-generating purposes and need to be prioritized mindful of sudden long-term changes to urban populations that are likely to continue, such as the migration of people from rural to peri-urban areas.

When targeting investments, one should prioritize areas with low household water consumption and those that are particularly food insecure. Focus on supporting existing household investments that enable clean and reliable water supplies close to the home, as this has greatest impact on raising water consumption. Focus also on household sanitation as a step towards citywide inclusive sanitation. Noting that education is one of the few assets that rural people can take when they move, prioritize improving WASH, including menstrual hygiene management in schools, as support to the education sector will have long-term returns.

Long-term food security is paramount for the social and economic development of Somalia. One must safeguard and increase the production of both staple foods and livestock fodder. Building resilience in rural Somalia requires investments that can reduce the impacts of droughts on nomadic and agro-pastoral communities. Water and land management practices, including soil conservation and flood management, are essential elements of all projects. New projects could emulate the Biyoole project model wherein sustainable land management is promoted around the construction of a water point. The securing of the rural economy and building resilience will reduce the forced displacement that comes with the repeated shocks that Somalia experiences, and water access is central in this resilience building. Finally, the livestock sector would benefit from securing all-year fodder, and there are gains to be made through water investments throughout the livestock value chain.



Livestock watering at a water point in Hudur Bakool. (Photo courtesy: Ahmed Mohamed Hassan)

Expanding access to improved water sources and, in particular, ensuring that sources are located close to the home, will reduce the time that women need to spend on water collection, freeing up their time for other activities and reducing their vulnerability to violence. Investment into girls' education is extremely important for the economic and human development of the country, and WASH investments can play an important part in enabling girls to remain in school. Women have important roles to play in water management, which need to be very well supported.

5.3 Priorities for Filling Information and Data Gaps

Gaps in data and information prevent analysis of how water scarcity constrains economic prospects for households, firms, and the national economy. While the 'Somalia Economics of Water' study produced and analyzed an enormous amount of data, there remains much to be done. The points below discuss priorities for starting to fill the information and data gaps.

Understanding the role of the urban periphery in relation to why goods are not coming in from the hinterlands and defining the constraints to this in relation to water is, furthermore, important, as is also understanding why Somalia is importing staple vegetables such as onions and tomatoes from Ethiopia.

While not a water sector issue directly, the expansion of enclosures creates conflict and threatens traditional livestock migration. The livestock sector, thus, also needs to be better understood and taken into consideration.

There is a need to ensure that investments prioritize the legal, political, cultural, and practical aspects of gender inequalities in Somalia as well as the collection of gender-disaggregated data.

It is also relevant to explore and introduce other technologies that are not found in Somalia. Manual well drilling is one of these technologies and the report recommends following the tried and tested model in

Box 1. Data Scarcity

During the study, the paucity of published and consolidated data led to the project having the added benefit of becoming a producer and analyzer of data. The lack of data was a major constraint throughout the study, and data constraints affect many of the areas and variables that would prove very insightful for an economic assessment of how water scarcity impacts on economic prospects, both at the macroeconomic and microeconomic levels. Data constraints are described in depth in Appendix 3 of the full report.

FCV contexts: undertake an analytical study to understand the development potential for manual well drilling, if there is potential, then prepare and implement a small scale pilot project operation and, if successful, take to scale in a large groundwater investment project; areas for intervention should be the areas without wadis and which are therefore unsuitable for sand dams. Projects of this nature will, in time, provide a network of emergency wells in drought-prone areas providing rural safety nets.

An infrastructure assessment to understand the current state of the existing irrigation infrastructure is needed. This will determine if it is possible, and feasible, to rehabilitate or construct new irrigation infrastructure. A study of potential for spate irrigation can be done at the same time. More important, however, is the need to reinvent the riverine breadbasket and a quick win in the short term will be the introduction of farmer-led irrigation development (FLID) where the costs of irrigation are four times less than traditional irrigation and far easier to implement.

Understanding urban water supply and sanitation is important. The 'Somalia Economics of Water' study has dug deeply into rural water supply and future analytical work is required to position the sector for advancements in urban water supply and sanitation. More information needs to be gathered on water



Elevated tank in Aleybadey. (Photo courtesy: Ahmed Mohamed Hassan)

service delivery for human consumption in cities and towns and for commercial and industrial users. Critical in this is deepening understanding of large-scale extraction and storage to facilitate the preparation of urban investment in water supply. Unpacking the role of water utilities is key and little is currently known about the basics such as contracting modalities, financing sources, governing policies, and fiscal risks.

5.4 Short- and Medium-Term Projects

Based on the premise of learning by doing and a recognition that, in Somalia, tangible projects facilitate intergovernmental coordination and sectoral oversight, the tables below provide a list of investment priorities based on the analysis undertaken by this 'Economics of Water' study. The priorities presented in Tables 2 and 3 correspond to the short term (0–2 years) and medium term (0–5 years), respectively. The priorities are disaggregated by rural, peri-urban, and urban; alignments with the particular NWRS flagship projects are given.

Table 2. List of Investment Priorities: Short Term (0–2 Years)

Suggested Formulation	Timeframe	Focus	Alignment with NWRS Flagship Project #
Strengthen resilience at the local level			
Investment in new small water systems/expansion, including innovations (for example, sand dams) coupled with appropriate governance and management arrangements	Short	Rural	3, 7, 8, 10, 12
Build a network of emergency wells in vulnerable localities	Short	Rural	8
Provide technical assistance/co-funding for soil and water conservation techniques in selected areas	Short	Rural	8, 7, 10
Provide technical assistance/co-funding for private and community investments in water	Short	Rural	4
Strengthen local community capacity (1): rehabilitation of watering facilities with support for the reestablishment and strengthening of nomadic communities' organization, including conflict adjudication	Short	Rural	3, 7, 12

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Table 2. List of Investment Priorities: Short Term (0–2 Years)

Suggested Formulation	Timeframe	Focus	Alignment with NWRS Flagship Project #
Improve data, information, and knowledge			
River flow assessments, based on a combination of past data and hydrological modeling utilizing remote sensing data (including upstream basin sections in Ethiopia)	Short	All	6, 7, 9, 11
Research studies/further diagnostics on: • Delivery models, sources and access for select urban areas as basis of taking decisions in future service delivery • Boosting year-round affordable good quality livestock feed • How to best support the promotion of water and soil conservation and water harvesting methods in drylands • How to support integrated land and water management practices, including soil conservation and flood management • Viable flood mitigation and drought resilience measures including forest stewardship • Climate-resilient water and sanitation technologies and management techniques These studies should be embedded in, and undertaken as part of, investment operations	Short	All	5, 7, 8, 9, 10, 12
Economic studies on sectoral water uses (mostly WSS, livestock, and agriculture) but also on new trends of energy conversion efforts and incipient industrial development	Short	All	7, 9, 10, 12
Assessment of the potential use of economic incentives for sustainable water management (mainly pricing mechanisms, risk management schemes, and cooperation-based instruments)	Short	All	5, 9
Undertake an in-depth diagnostic of the existing water supply and sanitation facilities in capital and secondary cities. The focus should be on clearly defined sources of water, and how to collect, store, and deliver	Short	Urban	7, 9
Urban Investment Plans (1): Deepen analysis of 23 most distressed towns in terms of water resources towns to determine growth potential of, and inequalities in, and define key urban water infrastructure investment priorities	Short	Urban	3, 7, 9
Urban Investment Plans (2): Prepare secondary city summary investment plans for Galmudug, South West State, Hirshabelle, and Jubbaland	Short	Urban	3, 7, 9

Table 3. List of Potential Investments for the Medium Term (0–5 Years)

Suggested Formulation	Timeframe	Focus	Alignment with NWRS Flagship Project #
Strengthen resilience at the local level			
Develop perennial livestock feed crops that utilize ‘green water’, but at times have blue river water added (supplemental irrigation). This would form the basis for high-quality livestock export (well-fed, healthy livestock, selling at good prices)	Medium	Rural	7, 12
Introduce consultative, participatory approaches whereby water access helps livestock stakeholders through participatory rangeland management and improves environmental stewardship of rangelands, including their adaptation to an increasingly threatening climate change	Medium	Rural	7, 12
Transforming livelihoods and the economy			
Large scale introduction of manual well drilling (training, local equipment production, private sector development)	Medium	All	5, 9
Promote farmer-led irrigation development	Medium	Rural	10
Provision of large volumes of water to enable the gradual commercialization of Jubba and Shabelle (river flow dependent) agriculture, from low-value crops to higher-value crops	Medium	Rural, Peri-urban	7, 10
Pilot locally based Natural Resources Management and Water Resources Management including soil rehabilitation	Medium	Rural	8, 10, 11
Connect urban and rural provision of water for households, which may entail interventions and deeper assessments as per the optimal scale for the provision of these services by specialized water utilities or other operators (in cooperation with community-based systems)	Medium	Rural	7, 9
Connect water supply and wastewater treatment, thus making the best out of the potential for reclaiming and reusing water	Medium	Rural	9
In coastal towns, pilot mixed source water provision through complementing fresh groundwater with desalination such as reverse osmosis options	Medium	Urban	9

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Table 3. List of Potential Investments for the Medium Term (0–5 Years)

Suggested Formulation	Timeframe	Focus	Alignment with NWRS Flagship Project #
Improve data, information and knowledge			
Deepen transboundary basin knowledge and understanding: - Establish a basis for transboundary basin management - Undertake situation assessment and basin development scenario studies in the Jubba and Shabelle basins - Undertake situation assessment and basin development scenario studies in transboundary aquifers - Develop principles and policy for transboundary cooperation and basin management - Promote trust building activities towards transboundary basin management across the region	Medium	Rural	2, 3, 5, 6
A policy intervention is needed relating to the defining of a clear urban service delivery model by government	Medium	Urban	9
Urban Investment Plans (3): Develop an overarching Urban Water Sector Investment Plan	Medium	Urban	4, 9



Bibliography

- Auerbach, Alan J., and L.J. Kotlikoff. 1987. *Dynamic Fiscal Policy*. Cambridge, U.K.: Cambridge University Press.
- Basnyat, D.B. 2007. *Water Resources of Somalia*. Technical Report No W-11. Nairobi, Kenya: FAO SWALIM (GCP/SOM/045/EC). https://www.faoswalim.org/resources/site_files/W-11%20Water%20Resources%20of%20Somalia_0.pdf
- Gaur M.K., and V.R. Squires. 2018. Geographic Extent and Characteristics of the World's Arid Zones and Their Peoples. In M. Gaur, V. Squires, eds., *Climate Variability Impacts on Land Use and Livelihoods in Drylands*. Springer, Cham. https://doi.org/10.1007/978-3-319-56681-8_1
- EarthWater. 1998. Pre-feasibility Geophysical investigations. Garowe Town Water Supply. Puntland State, Somalia.
- FAO, AQUASTAT. 2014. Country profile Somalia. Retrieved January 8, 2021, from <http://www.fao.org/aquastat/en/countries-and-basins/country-profiles/country/SOM>
- FAO and WB (Food and Agriculture Organization and the World Bank). 2018. *Rebuilding Resilient and Sustainable Agriculture in Somalia*. Washington, D.C. <http://documents.worldbank.org/curated/en/803231522165074948/Volume-1-Overview>
- FGS (Federal Government of Somalia). 2018. *Drought Impact and Needs Assessment*. Mogadishu, Somalia.
- Khandker V., V.P. Gandhi, and N. Johnson. 2020. "Gender Perspective in Water Management: The Involvement of Women in Participatory Water Institutions of Eastern India". *Water* 12(1):196. <https://doi.org/10.3390/w12010196>
- Liniger, H., and W. Critchley. 2007. Where the land is greener: Case-studies and analysis of soil and water conservation initiatives worldwide. CTA/CDE/FAO/UNEP/WOCAT.
- Louis Berger Inc. 1985. Comprehensive ground-water development, project 104. Final report Volume II, Hydrology. Somali Democratic Republic, Ministry of Minerals and Water resources, Water Development Agency.
- Lundqvist, J. 1999. *Mainstreaming Gender in Water Resources Management: Why and how*. Background Paper for the World Vision Process. Accessed May 7, 2021. Available at <http://www.pacificwater.org/userfiles/file/IWRM/Toolboxes/gender/MAINSTREAMING%20GENDER%20IN%20WATER%20RESOURCES.pdf>
- MoEWR. 2021. *Water Resources Strategy and Plan*. Ministry of Energy and Water Resources, Federal Government of Somalia.
- National Bureau of Statistics. 2021. *Somalia Gross Domestic Product Report*. Mogadishu, Somalia. <https://www.nbs.gov.so/gross-domestic-product-gdp/>
- Petersen, G., and H.M. Gadain. 2012. *Water Demand Assessment for the Juba and Shabelle Rivers*. Technical Report No W-22, FAO-SWALIM (GCP/SOM/049/EC) Project, Nairobi, Kenya. https://www.faoswalim.org/resources/site_files/W%2022%20Water%20Demand%20Assessment_0.pdf
- Rubel, F., and M. Kottek. 2010. "Observed and projected climate shifts 1901–2100 depicted by world maps of the Köppen-Geiger climate classification." *Meteorologische Zeitschrift* 19(2): 135–141.
- Said, A.A., R. Yurtal, M. Cetin, and M.S. Gölpinar. 2021. "Evaluation of some groundwater quality parameters using geostatistics in the urban coastal aquifer of Bosaso plain, Somalia." *Journal of Agricultural Sciences*. <https://dergipark.org.tr/en/download/article-file/794340>
- Shilenje, Z.W., and V. Ongoma. 2014. "Observed surface ozone trend in the year 2012 over Nairobi, Kenya." *Atmósfera* 27(4): 377–384.

- Tierney, J.E., C.C. Ummenhofer, and P.B. de Menocal. 2015. "Past and future rainfall in the Horn of Africa." *Sci. Adv.* 1, e1500682. https://www.researchgate.net/publication/283851190_Past_and_future_rainfall_in_the_Horn_of_Africa
- UNICEF. 2011. MICS Somaliland, 2011.
- van Haren, I., L. Bastidas, N. Limones, and M. Wijnen. 2017. Assessment of Water Resources for Priority Areas in Somaliland. Final Report. Somalia Drought Emergency Response. Water Partnership Program, World Bank Group.
- van Wijk-Sijbesma, C. 1998. Gender in Water Resources Management, Water Supply and Sanitation: Roles and Realities Revisited. International Research Centre for Water and Sanitation. Delft, The Netherlands.
- WASH Cluster Somalia. Draft 2017. Minimum WASH Technical Guidelines. 180502_wash_technical_guidelines_2017.pdf (humanitarianresponse.info)
- WHO and UNICEF. 2019. Progress on Drinking Water, Sanitation and Hygiene: 2017 Update and SDG Baseline. WHO & UNICEF, Joint Monitoring Programme, Geneva and New York.
- ———. 2021. Estimates for drinking water, sanitation and hygiene services by country (2000–2020). Updated July 2021. Accessed July 15, 2021. www.washdata.org
- World Bank. 2018. Somalia Drought Impact and Needs Assessment. Volume II: Sector reports.
- ———. 2021a. Somalia Urbanization Review: Fostering Cities as Anchors of Development. Washington, D.C. <https://openknowledge.worldbank.org/handle/10986/35059>
- ———. 2021b. Horn of Africa Regional Economic Memorandum: Overview Report. Washington, D.C.
- ———. 2021c. Gender Gaps in Somalia's Water Sector. Washington, D.C.
- World Bank Group. 2020. Domestic Resource Mobilization in Somalia. Washington, D.C. <http://documents.worldbank.org/curated/en/644541596773094934/Domestic-Resource-Mobilization-in-Somalia>
- ———. 2021a. Improving Access to Jobs for the Poor and Vulnerable in Somalia. Washington, D.C. <http://documents.worldbank.org/curated/en/540261609935995160/Improving-Access-to-Jobs-for-the-Poor-and-Vulnerable-in-Somalia>
- ———. 2021b. Trade as an Engine of Growth in Somalia: Constraints and Opportunities. Washington, D.C.
- ———. 2021c. Somalia Country Economic Memorandum: Towards an Inclusive Jobs Agenda. Washington, D.C. <http://documents.worldbank.org/curated/en/464291625747249500/Somalia-Economic-Memorandum-Towards-an-Inclusive-Jobs-Agenda>





Notes

